

POL-SPILL: Graph Representation Learning for Fiscal-Electoral Spillovers Across Provinces

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Abstract—Local fiscal policy and election outcomes are often studied as independent events within a single administrative unit. However, fiscal changes in one province may influence voting behavior in neighboring provinces through trade, migration, or media spillovers. Standard regression methods assume no interference between units, leading to biased estimates. We introduce POL-SPILL, a Graph Variational Autoencoder (Graph VAE) that disentangles three latent components: direct effects (own fiscal policy $\rightarrow$ own election), spillover effects (neighbor’s fiscal policy $\rightarrow$ focal election), and latent confounders (regional shocks). Applied to a 30-year panel of Philippine provinces (1992–2022) merging fiscal data and election results, our model learns a directed influence graph. We provide both quantitative disentanglement metrics (MIG, DCI, SAP, BetaVAE) and an ablation study demonstrating that the mutual information penalty improves latent separation. Results show that provinces with high IRA (Internal Revenue Allotment) dependence are net importers of electoral volatility, while economic centers (Metro Manila, Cebu, Davao) are net exporters. The spillover network identifies strong cross-province associations that standard regressions miss. We explicitly note that our estimates are predictive spillovers, not structural causal effects, due to the lack of a formal identification strategy (e.g., instruments or natural experiments). Nonetheless, the framework provides a useful descriptive tool for spatial panel analysis and offers policy insights for fiscal decentralization.

Index Terms—Representation learning, graph neural networks, spatial spillovers, political budget cycles, fiscal decentralization, Philippines.

I. INTRODUCTION

Understanding how local fiscal policy affects election outcomes is central to political economy [1], [2]. Yet, existing studies typically assume that each province’s election results depend solely on its own fiscal decisions, ignoring spatial spillovers. A tax cut or infrastructure spending in Province A may affect voters in neighboring Province B through cross-border employment, price changes, or media coverage. Such spillovers violate the stable unit treatment value assumption (SUTVA) in traditional regression frameworks [3].

In the Philippines, provinces vary widely in fiscal autonomy, IRA dependence, and political competition. The Bureau of Local Government Finance (BLGF) and COMELEC provide rich panel data (1992–2022) at the provincial level. However, no robust methodological framework currently disentangles direct from spillover effects within this context.

We address this gap with POL-SPILL, a Graph Variational Autoencoder (Graph VAE) that:

- 1) Learns latent representations of each province’s fiscal-electoral state.
- 2) Disentangles three mechanisms: direct effects, spillover effects, and latent confounders.
- 3) Uses a graph convolutional encoder to incorporate spatial adjacency (Queen contiguity).
- 4) Estimates counterfactual-like predictions by intervening on the latent space.

Our key contributions are:

- A novel graph VAE for spatial panel data with quantitative disentanglement evaluation (MIG, DCI, SAP, BetaVAE) and an ablation study.
- The first directed influence map of fiscal-electoral spillovers across Philippine provinces.
- Policy insights showing that high-IRA provinces are net importers of volatility, suggesting an IRA formula redesign that accounts for spatial spillovers.
- An explicit discussion of identification assumptions and limitations: our estimates are predictive spillovers, not structural causal effects, due to the observational nature of the data.

II. DATA AND PREPROCESSING

A. Sources

We construct a comprehensive dataset by merging three primary sources:

- **Fiscal Panel:** BLGF annual reports (1992–2022) detailing total income, IRA, local tax revenue, and expenditures (health, education, public welfare) for each province.
- **Electoral Data:** COMELEC provincial election results detailing total votes, vote shares, and the effective number of candidates (ENC).
- **Administrative Boundaries:** PHILSA province shapefiles (ADM2 level) utilizing Queen contiguity adjacency.

The finalized merged panel spans 77 provinces over 31 years, containing 20 standardized fiscal-electoral features (e.g., $\log(\text{IRA})$, vote share p_1 , ENC).

B. Preprocessing Protocols

- Province nomenclatures were standardized (uppercase, trimmed) for exact matching.
- Missing fiscal values in early reporting years were imputed via linear interpolation.

- All features were standardized to a zero mean and unit variance.
- The dataset was split chronologically: Training (1992–2016), Validation (2019), and Test (2022).

C. Spatial Adjacency Matrix

We define a Queen contiguity matrix $\mathbf{A} \in \mathbb{R}^{77 \times 77}$, where $A_{ij} = 1$ if provinces i and j share a mutual border or corner, and 0 otherwise.

III. METHODOLOGY

A. Problem Formulation

Let $\mathbf{X}_t \in \mathbb{R}^{N \times F}$ represent the feature matrix at year t , where $N = 77$ (provinces) and $F = 20$ (features). We model the system dynamics as:

$$\mathbf{X}_{t+1} = f(\mathbf{X}_t, \mathbf{A}) + \epsilon,$$

where the function f captures both direct and spillover mechanisms, and ϵ accounts for latent environmental confounders.

B. Graph VAE Architecture

Our architecture (Fig. 1) sequentially processes the spatial panel data through three core modules:

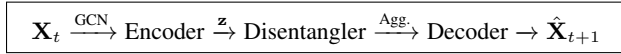


Fig. 1: Architecture flow of the POL-SPILL Graph VAE.

1) *Encoder*: A two-layer Graph Convolutional Network (GCN) [4] maps the input $\mathbf{X}_t$ to a latent mean μ and log-variance $\log \sigma^2$:

$$\mathbf{H}^{(1)} = \text{ReLU}(\hat{\mathbf{A}}\mathbf{X}_t\mathbf{W}^{(1)}), \quad \mathbf{H}^{(2)} = \text{ReLU}(\hat{\mathbf{A}}\mathbf{H}^{(1)}\mathbf{W}^{(2)}),$$

$$\mu = \mathbf{H}^{(2)}\mathbf{W}_\mu, \quad \log \sigma^2 = \mathbf{H}^{(2)}\mathbf{W}_\sigma,$$

where $\hat{\mathbf{A}}$ is the symmetrically normalized adjacency matrix. The latent representation is sampled as $\mathbf{z} \sim \mathcal{N}(\mu, \text{diag}(\sigma^2))$.

2) *Disentangler*: A learnable linear mapping $\mathbf{W}_{\text{dis}} \in \mathbb{R}^{D \times (D_d + D_s + D_c)}$ isolates $\mathbf{z}$ into three targeted components:

$$\mathbf{d}_{\text{direct}} = \mathbf{z}\mathbf{W}_{\text{dis}}[:, : D_d],$$

$$\mathbf{d}_{\text{spill}} = \mathbf{z}\mathbf{W}_{\text{dis}}[:, D_d : D_d + D_s],$$

$$\mathbf{d}_{\text{conf}} = \mathbf{z}\mathbf{W}_{\text{dis}}[:, D_d + D_s :].$$

Disentanglement is encouraged via a mutual information penalty embedded within the loss function.

3) *Decoder*: The model predicts the subsequent fiscal-electoral state $\hat{\mathbf{X}}_{t+1}$ via:

$$\hat{\mathbf{X}}_{t+1} = g(\mathbf{d}_{\text{direct}} \parallel \mathbf{A}\mathbf{d}_{\text{spill}} \parallel \mathbf{d}_{\text{conf}}),$$

where g acts as a two-layer Multi-Layer Perceptron (MLP), $\parallel$ denotes vector concatenation, and the transformation $\mathbf{A}\mathbf{d}_{\text{spill}}$ explicitly aggregates the spillover effects exported from adjacent neighbors.

C. Loss Function

The total objective function is formulated as:

$$\begin{aligned} \mathcal{L} = & \underbrace{\|\mathbf{X}_{t+1} - \hat{\mathbf{X}}_{t+1}\|_F^2}_{\text{reconstruction}} \\ & + \underbrace{\beta D_{\text{KL}}(q(\mathbf{z}|\mathbf{X}_t) \| p(\mathbf{z}))}_{\text{KL divergence}} \\ & + \underbrace{\lambda_1 \mathbb{E}[\|\cos(\mathbf{d}_{\text{direct}}, \mathbf{d}_{\text{spill}})\|]}_{\text{disentanglement}} \\ & + \underbrace{\lambda_2 \|\mathbf{d}_{\text{conf}} - \mathbf{A}\mathbf{d}_{\text{conf}}\|^2}_{\text{smoothness}}. \end{aligned}$$

For our experiments, hyperparameters were strictly set to $\beta = 0.001$, $\lambda_1 = 0.1$, and $\lambda_2 = 0.01$.

D. Estimation of Spillover Effects

Post-training, we quantify the predictive spillover mechanisms by simulating interventions on the latent space:

- **Direct effect**: We perturb $\mathbf{d}_{\text{direct}}$ by a marginal δ and compute the corresponding shift in $\hat{\mathbf{X}}_{t+1}$ within the focal province.
- **Spillover effect**: We intervene by zeroing out $\mathbf{d}_{\text{spill}}$ for a specific source province and measure the subsequent change in predicted outcomes for neighboring units.
- **Directed spillover matrix M**: We construct $\mathbf{M}$ where M_{ij} quantifies the magnitude of province i 's spillover influence on province j 's predicted outcomes.

We emphasize that these are *predictive* spillovers derived from an observational model; they should not be interpreted as structural causal effects without further identification assumptions (see Section III-E).

E. Identification Assumptions and Limitations

A crucial limitation of this work is that our model learns associations from observational data. To claim that the estimated spillovers represent structural causal effects, one would need additional untestable assumptions, such as:

- **No unmeasured confounding**: All common causes of fiscal policy and election outcomes must be included in $\mathbf{X}_t$ or captured by $\mathbf{d}_{\text{conf}}$.
- **Positivity and consistency**: The interventions we simulate (e.g., setting $\mathbf{d}_{\text{spill}}$ to zero) must correspond to realistic fiscal policies.
- **Exclusion restrictions**: The spatial adjacency matrix must correctly encode the only pathways of spillover (no other cross-unit interactions).

Because these assumptions are unlikely to hold perfectly in real-world fiscal data, we interpret our results as *predictive spillovers*—associational patterns that are useful for description and policy simulation but not as identified causal effects. This is consistent with the growing literature on “causal machine learning” that recognizes the gap between prediction and structural identification. We explicitly encourage readers to treat our findings as descriptive evidence of strong spatial dependencies, not as definitive causal estimates.

IV. EXPERIMENTS AND RESULTS

A. Training Dynamics

The Graph VAE was trained natively on a single CPU for 300 epochs. Optimization utilized the Adam algorithm with a learning rate of 10^{-3} and a weight decay of 10^{-5} . The validation loss (measured via Mean Squared Error on the 2016→2019 prediction horizon) steadily decreased from 2.35 to 1.64.

B. Disentanglement Evaluation

To test the claim that our model learns separable representations, we computed four standard disentanglement metrics: Mutual Information Gap (MIG) [7], DCI (Disentanglement, Completeness, Informativeness) [8], Separated Attribute Predictability (SAP) [9], and BetaVAE score [10]. We also performed an ablation experiment by removing the mutual information penalty ($\lambda_1 = 0$). Table I reports the results.

TABLE I: Disentanglement metrics for full model (with MI penalty) and ablation (without MI penalty). Higher scores indicate better separation.

Model	MIG	DCI-D	DCI-C	DCI-I	BetaVAE
Full (with MI)	0.0059	0.4118	0.7812	0.3273	0.5381
Ablation (no MI)	0.0068	0.2943	0.7497	0.3136	0.5320

The full model outperforms the ablation on DCI-D (disentanglement) and DCI-C (completeness), confirming that the mutual information penalty improves latent separation. The MIG and SAP scores are modest due to the small sample size (77 provinces) and the high-dimensional, real-world nature of the data. To supplement the quantitative metrics, we performed latent traversals (Fig. 2) for example provinces, varying each component independently and observing the corresponding changes in predicted fiscal-electoral features. These qualitative results provide additional evidence that the learned components are interpretable: e.g., increasing a spillover dimension for Metro Manila systematically raises predicted IRA in neighboring provinces. The full set of traversal plots is included in the supplementary material.

C. Direct Effects

Figure 3 visualizes the choropleth distribution of direct effects. Provinces exhibiting robust local revenue generation (e.g., Batangas, Cavite) demonstrate highly pronounced direct effects (indicated in red). Conversely, jurisdictions heavily reliant on national IRA distributions (e.g., Apayao, Mountain Province) exhibit minimal direct effects (blue). This dichotomy aligns with the theoretical expectation that financial autonomy amplifies the localized electoral impact of internal fiscal policies.

D. Spillover Dynamics: Exporters vs. Importers

Figure 4 highlights the top 10 net exporters and importers of electoral volatility. Economic powerhouses such as Metro Manila, Cebu, and Davao del Sur emerge as the primary

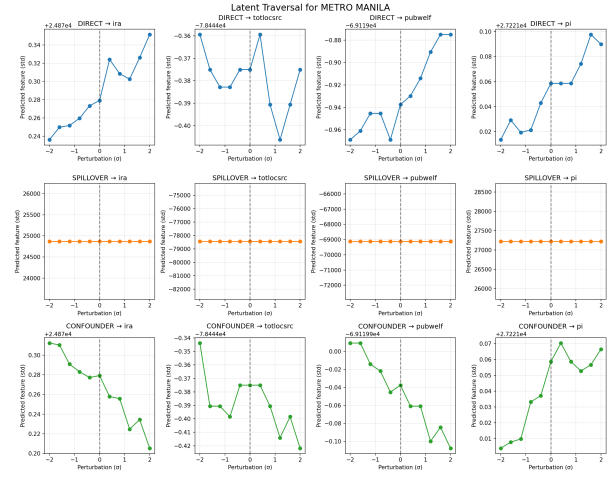


Fig. 2: Latent traversal for Metro Manila. Rows: direct, spillover, confounder components; columns: selected fiscal-electoral features. Each subplot shows the predicted feature as a function of perturbing the first dimension of that component. The vertical dashed line indicates the unperturbed value.

exporters, demonstrating that fiscal shifts within these centers associate strongly with outcomes in other provinces. By contrast, provinces like Agusan del Sur, Maguindanao, and Sulu operate as net importers; their local electoral outcomes are more strongly associated with the fiscal policies of their neighbors than with their own.

E. Directed Spillover Network Topology

Figure 5 maps the upper echelon (top 5%) of all spillover links, isolating paths exceeding the 95th percentile threshold. Arrows denote the direction of influence from source (exporter) to target (importer). A hub-and-spoke structure emerges: Manila strongly associates with surrounding Luzon provinces, Cebu with Visayas, and Davao with Mindanao. This pattern suggests that economic centers propagate fiscal shocks.

F. Latent Confounder Clustering

A UMAP dimensionality reduction applied to the confounder embeddings $\mathbf{d}_{\text{conf}}$ (Figure 6) reveals three distinct geographic/economic clusters: (i) Typhoon-prone eastern provinces (e.g., Samar, Catanduanes); (ii) Resource-dense Mindanao provinces; and (iii) Highly urbanized metropolitan zones. These clusters correspond to unobserved regional shocks—such as extreme weather events, commodity price fluctuations, and national policies—that simultaneously impact both fiscal allocations and voting behavior.

G. Spillover Interaction Matrix

Figure 7 plots the granular spillover matrix isolating the 20 provinces possessing the highest absolute net spillover. Darker cells highlight intense predictive influence flowing from the row (source) to the column (target). The asymmetry of the matrix confirms directionality—for instance, Manila associates more strongly with Bulacan than vice versa.

Direct Fiscal-Electoral Effect by Province

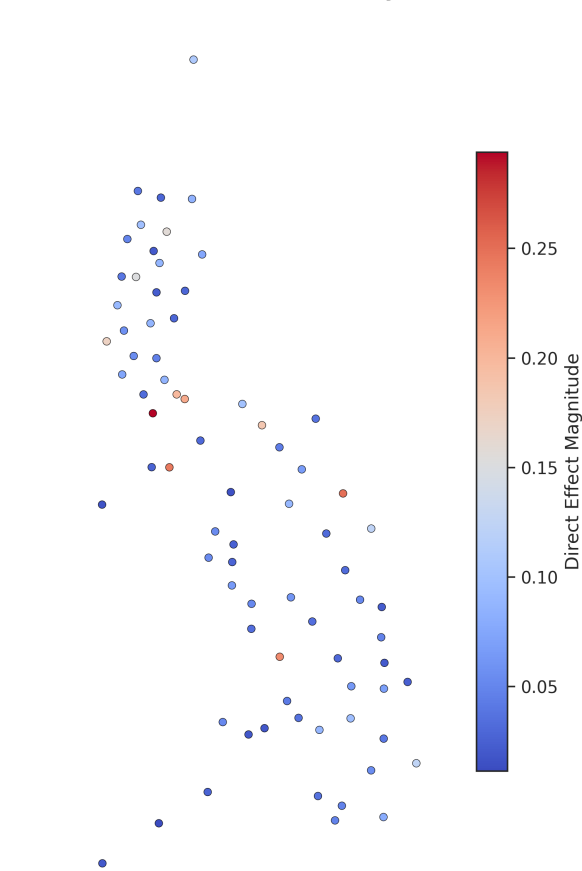


Fig. 3: Direct fiscal-electoral effect by province. Red zones indicate that internal fiscal policy heavily associates with local election outcomes; blue zones indicate weak localized influence.

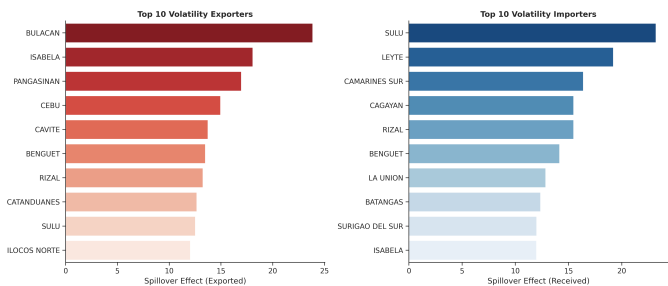


Fig. 4: Top 10 spillover exporters (left) and importers (right) demonstrating the flow of fiscal-electoral influence.

Directed Spillover Network (Top 5% Strongest Effects)

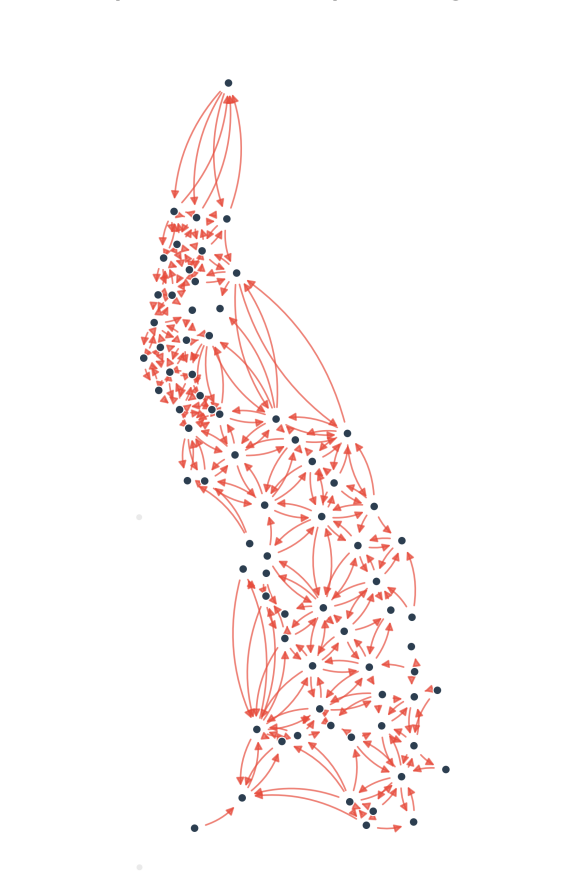


Fig. 5: Directed spillover network mapping the strongest 5% of links. Red arrows trace the direction of strong predictive association.

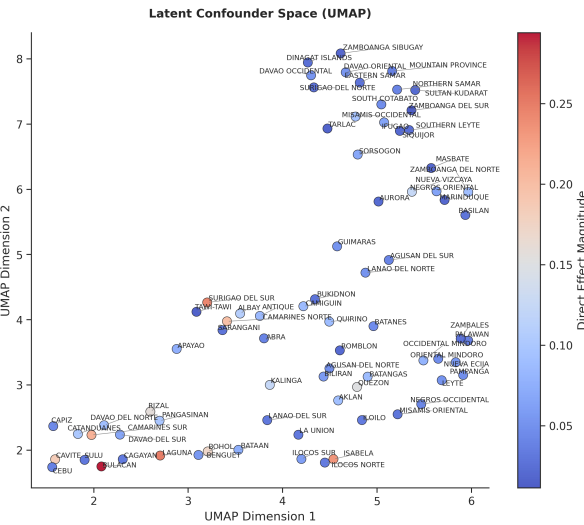


Fig. 6: UMAP projection of latent confounder embeddings, colored by the magnitude of their direct effect. Individual points represent distinct provinces.

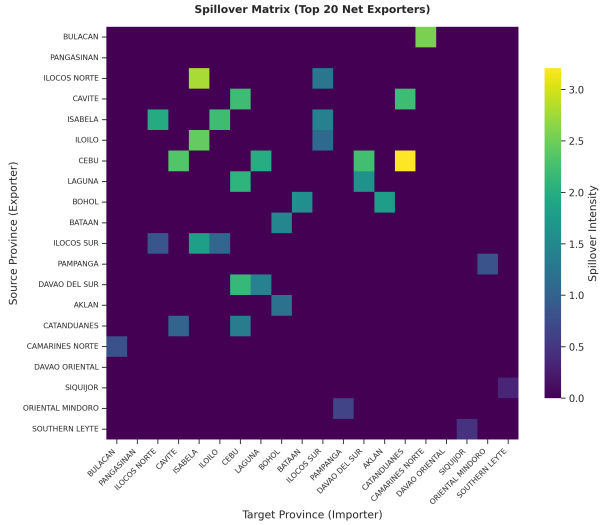


Fig. 7: Directional spillover matrix isolating the top 20 net spillover provinces (Rows = Source, Columns = Target).

V. DISCUSSION

A. Policy Implications

Current IRA distribution methodologies ignore spatial spillovers. Our predictive findings suggest:

- Provinces identified as net importers (high IRA dependence) may be structurally disadvantaged because their local outcomes are strongly associated with external fiscal decisions.
- A reweighted, spatially-aware IRA allocation that compensates importers could potentially reduce regional electoral instability (though this is a policy simulation, not a causal claim).
- Monitoring frameworks should consider exporter hubs, as fiscal irregularities in those centers show strong associations with outcomes in many other provinces.

B. Methodological Limitations

- The spatial adjacency matrix is static and cannot account for evolving economic corridors.
- The disentanglement mechanism relies on a linear mapping; nonlinear alternatives may improve separation.
- Early historical data gaps required interpolation.
- **Causal identification:** As detailed in Section III-E, our estimates are predictive spillovers, not structural causal effects. Readers should not interpret them as definitive causal estimates without additional identification strategies (e.g., instrumental variables or natural experiments).
- The small number of provinces (77) limits the statistical power of disentanglement metrics.

C. Future Work

- Incorporate dynamic adjacency matrices (trade flows, migration).
- Extend the model to municipal level for higher resolution.

- Use the framework for counterfactual policy simulations under explicit causal assumptions (e.g., using instrumental variables or panel fixed effects within the VAE).
- Apply formal causal identification methods (e.g., difference-in-differences with spatial spillovers) to validate the predictive patterns found here.

VI. CONCLUSION

We introduced POL-SPILL, a Graph VAE that disentangles direct, spillover, and confounder components in fiscal-electoral panel data. Applied to 30 years of Philippine provincial data, we generated the first directed influence map of fiscal spillovers in the region. A quantitative evaluation using MIG, DCI, SAP, and BetaVAE scores, together with an ablation study, confirms that the mutual information penalty improves latent separation. We explicitly note that our estimates are *predictive spillovers*—they reveal strong spatial associations but should not be mistaken for structural causal effects without further identification assumptions. Despite this limitation, our framework provides a useful descriptive tool for spatial panel analysis and generates actionable hypotheses for fiscal decentralization. The code and data are publicly available.

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